CO2 - AT2 - Part II

APPLICATION BASED QUESTIONS - TD (0), SARSA AND Q-LEARNING

TOPIC: DIGITAL FORENSICS

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APPLICATION 1: TD (0) ALGORITHM FOR DIGITAL FORENSICS

Proposed Algorithm for TD (0)

Step 1: Define States and Rewards

Let

- **S** = Set of forensic investigation states
- **A** = Set of possible investigator actions
- **R** = Immediate reward
- **γ** = Discount factor
- **α** = Learning rate

Step 2: Initialize Value Function

Initialize

$$V(s)=0 \quad (1)$$

Eq. (1): Initializes the value of every forensic state before learning begins.

Step 3: Observe Current State

Agent observes current forensic state

$$st \quad (2)$$

Eq. (2): Represents the present investigation state from which learning starts.

Step 4: Receive Reward and Next State

After taking an action,

$$(st, rt + 1, st + 1)(3)$$

Eq. (3): The system receives an immediate reward and moves to the next investigation state.

Step 5: TD Error Calculation

$$\delta = r + \gamma V(s') - V(s)(4)$$

Eq. (4): Computes the Temporal Difference (TD) error by comparing predicted and observed values.

Step 6: Update State Value

$$V(s) = V(s) + \alpha\delta(5)$$

Eq. (5): Updates the value of the current forensic state using the TD error.

Step 7: Prediction

Final learned values identify the most valuable investigation states.

Algorithm 1: TD (0)

1. Initialize V(s)=0
2. Observe current state
3. Take action
4. Receive reward and next state

5. Compute TD error

6. Update V(s)

7. Repeat until convergence

Sample Calculation

Given

$\alpha = 0.2$

$\gamma = 0.9$

Reward = 10

V(s)=20

V(s')=25

Step 1

TD Error

$\delta=10+0.9(25)-20$

$=10+22.5-20$

$=12.5$

Step 2

Updated Value

$V(s)=20+0.2(12.5)$

$=20+2.5$

$=22.5$

Prediction

The updated forensic state value becomes **22.5**, indicating improved confidence in the investigation path.

APPLICATION 2: SARSA ALGORITHM FOR DIGITAL FORENSICS

Proposed Algorithm for SARSA

Step 1: Initialize Q-table

$$Q(s,a) = 0 \quad (6)$$

Eq. (6): Initializes all state-action values to zero.

Step 2: Select Action

Choose action using ε-greedy policy.

$$a = \pi(s) \quad (7)$$

Eq. (7): Selects an action balancing exploration and exploitation.

Step 3: Receive Reward

Observe

$$(r, s', a') \quad (8)$$

Eq. (8): Receives reward and selects the next action.

Step 4: SARSA Update

$$Q(s,a) = Q(s,a) + \alpha[r + \gamma Q(s',a') - Q(s,a)] \quad (9)$$

Eq. (9): Updates the Q-value using the action actually taken in the next state.

Step 5: Prediction

The learned Q-table recommends the safest forensic investigation action.

Pseudocode

Algorithm 2 : SARSA

1. Initialize Q-table
2. Select action
3. Execute action
4. Receive reward
5. Select next action
6. Update Q-value
7. Repeat

Sample Calculation

Given

α=0.2

γ=0.9

Reward=15

Q(s,a)=12

Q(s',a')=18

Step 1

Target

15+0.9(18)=31.2

Step 2

Update

Q=12+0.2(31.2−12)

=12+3.84

=15.84

Prediction

The optimal forensic action value becomes **15.84**.

APPLICATION 3: Q-LEARNING FOR DIGITAL FORENSICS

Proposed Algorithm for Q-Learning

Step 1: Initialize Q-table

$$Q(s,a) = 0 \quad (10)$$

Eq. (10): Initializes all Q-values before learning.

Step 2: Execute Action

Observe reward and next state.

$$(s, a, r, s')(11)$$

Eq. (11): Executes an action and records the resulting transition.

Step 3: Find Best Future Value

$$\max Q(s', a') \ (12)$$

Eq. (12): Finds the maximum future action value for the next state.

Step 4: Update Q-value

$$Q(s, a) = Q(s, a) + \alpha[r + \gamma \max Q(s', a') - Q(s, a)](13)$$

Eq. (13): Updates the Q-value using the maximum future reward.

Step 5: Prediction

The optimal forensic investigation policy is obtained from the learned Q-table.

Pseudocode

Algorithm 3 : Q-Learning

1. Initialize Q-table
2. Observe state
3. Select action
4. Receive reward
5. Find maximum future Q-value

6. Update Q-value
7. Repeat until convergence

Sample Calculation

Given

$\alpha=0.2$

$\gamma=0.9$

Reward=20

Current Q=18

Maximum Future Q=30

Step 1

Target

$20+0.9(30)=4$

Step 2

Update

$Q=18+0.2(47-18)$

$=18+5.8$

$=23.8$

Prediction

The optimal investigation action value becomes **23.8**, indicating the best action for future digital forensic investigations.